# DevSync — Test Plan

Secure AI Project Collaborator · Graduation Project (DEPI) · Flutter + Firebase · Android

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## 1. Objective and Scope

### 1.1 Objective

Make sure all approved features work correctly on real devices and in the release build, and that the KPIs and NFRs are met before the final delivery (4 December 2026).

### 1.2 In Scope

- 13 modules (about 64 screens and 147 operations) according to the Core Features document: Splash/Onboarding, Authentication (with Guest), Profile, GitHub Integration, AI Match, Project Discovery, Community, Project Details, Create Project, My Teams, My Projects, Prompt Evaluation, Chatbot.
- Both the Free and Pro flavors (the feature flag and the upgrade information screen).
- Non-functional requirements NFR-01 to NFR-14.

### 1.3 Out of Scope

iOS, real payment processing, full Adaptive Matching (the app learning from rejections), AI analysis of the CV, CI and issue tracker integration, and the Digital Twin as a standalone feature.

### 1.4 Note on Priorities

All Must items have to be tested. Should/Could items (Community, Chat, Chatbot, AI Project Assistance, Team Invite/Shared link) are tested only if they are implemented, and they are marked (S/C) in the tables.

## 2. Test Levels and Environment

### 2.1 Levels

| Level | What we test | Tool |
|---|---|---|
| Unit | MatchingService (8 to 10 cases: full match, zero match, incomplete profile, duplicate skills), validators, Cubits, prompt version selection | flutter_test, bloc_test |
| Integration | Auth with Firestore, Cloud Functions (GitHub and LLM), FCM, Firestore/Storage security rules | Firebase Emulator + a test project |
| System / E2E | The full flow on two real devices with the release build | Manual, following the Test Cases |
| UAT | 10 testers (colleagues) + a feedback form to measure the KPIs | Google Form |
| Non-functional | Performance, security, compatibility, stability | Timing measurements, Crashlytics, manual review |

### 2.2 Environment

- At least two real Android devices with different versions and screen sizes (NFR-12).
- Release build (AAB and APK with obfuscation) for both the free and pro flavors.
- A Firebase test project + Emulator (we need the Blaze plan for Cloud Functions).
- Seed data: 15 to 20 projects (Learning and Real, with different difficulties, technologies and roles, and some without a Backend or QA role) + 5 to 10 test profiles, loaded with a single command.

### 2.3 Test Accounts

| Account | Description |
|---|---|
| Guest | No account |
| Dev-A | Complete profile + GitHub connected |
| Dev-B | Incomplete profile and no GitHub |
| Owner | Owner of a published project |
| Member | Member of a team |
| Applicant | Someone who submitted a join request |
| Tester-01 to 10 | UAT testers |

## 3. Criteria and Procedures

### 3.1 Severity Levels

| Level | Definition |
|---|---|
| Critical | A crash, the core flow is blocked, or a data leak |
| High | A Must feature does not work but there is a workaround |
| Medium | Wrong behavior that does not block work |
| Low | Visual or text issue |

### 3.2 Entry Criteria

- The feature has been merged into develop and works on a real device.
- The seed data is ready.
- The release trial build is ready (before 20 November).

### 3.3 Exit Criteria

- 100% of the Must Test Cases have a result.
- Zero open Critical and zero open High bugs.
- Every bug has a GitHub Issue.
- The KPIs have been measured (section 7).
- The full flow has been tried on two devices.

### 3.4 Bug Report Template (GitHub Issues)

Title · Steps · Expected · Actual · Device and Android version · Build (free/pro and debug/release) · Severity · Screenshot or video.

### 3.5 Schedule

| Date | Activity |
|---|---|
| 6 November | Test Plan delivery |
| 16 to 30 November | Logging Bug Reports on GitHub Issues |
| 19 to 20 November | QA Round 1: the full flow on two devices with the release build |
| 20 to 28 November | KPI measurement with 10 testers |
| 24 to 28 November | QA Round 2: regression after the fixes |
| 30 November | Delivery of Test Results and Bug Reports |
| 4 December | Final presentation and reports |

## 4. Test Cases by Module

Symbols: Priority M = Must, S/C = Should/Could. Type: U = Unit, I = Integration, S = System/UI, Sec = Security. Result: ☐ not yet run, ✔ passed, ✘ failed.

### 4.1 Splash and Onboarding (6 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **ONB-01** | Check for a previous session | Open the app while logged in | Splash, then straight to Discover without Onboarding | M/S | ☐ |
| **ONB-02** | Check the first-launch flag | Open the app for the first time, close it, open it again | Onboarding appears only once (SharedPreferences) | M/S | ☐ |
| **ONB-03** | Next and swipe between slides | Swipe or tap Next through the 3 slides | Navigation works correctly and the last slide has Get Started and Continue as Guest | M/S | ☐ |
| **ONB-04** | Skip | Tap Skip from any slide | Goes to the Auth Entry screen | M/S | ☐ |
| **ONB-05** | Get Started | Tap Get Started on the last slide | Opens Auth Entry | M/S | ☐ |
| **ONB-06** | Continue as Guest | Tap Continue as Guest | Enters Discover in read-only mode | M/S | ☐ |
| **ONB-07** | Animations | Watch the slide transitions (AnimatedContainer and Hero) | Smooth transitions with no stutter | M/S | ☐ |

### 4.2 Authentication (with Guest Mode) (10 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **AUTH-01** | Sign up with email | Enter name, email, password and confirmation, then Sign up | The account is created and goes to profile setup | M/I | ☐ |
| **AUTH-02** | Sign-up validation | Leave fields empty, enter a wrong email, a weak password, a mismatched confirmation | A clear error message under each field and nothing is submitted | M/U | ☐ |
| **AUTH-03** | Duplicate email | Sign up with an existing email | A clear message that the email is already in use | M/I | ☐ |
| **AUTH-04** | Send email verification | After signing up, open the email | The verification message arrived, and resend works | M/I | ☐ |
| **AUTH-05** | Log in with email | Enter correct credentials | Enters the account and goes to Discover or profile setup | M/I | ☐ |
| **AUTH-06** | Log in with wrong credentials | Wrong password or a non-existent email | A clear error message with no crash | M/S | ☐ |
| **AUTH-07** | Sign in with Google | Tap Continue with Google on the release build for each flavor | Sign-in succeeds (SHA-1 and SHA-256 added for each flavor) | M/I | ☐ |
| **AUTH-08** | Forgot password | Enter the email, then Send | The Check your inbox screen and the reset link arrived | M/I | ☐ |

| | | | | | |
|---|---|---|---|---|---|
| **AUTH-09** | Resend the reset link | Tap Resend | A new link arrives | M/I | ☐ |
| **AUTH-10** | Guest browsing | Enter as a Guest and open Discover, Project Details, Community Feed, Public profiles and the filters | All are available and read-only | M/S | ☐ |
| **AUTH-11** | Login gate | As a Guest, tap Join, Request to Join, Save, Express Interest, Create Project, Messages, Notifications, AI Match, Chatbot or Profile | The Login Required bottom sheet appears | M/S | ☐ |
| **AUTH-12** | Return to the original screen | After AUTH-11, log in | Returns to the same screen the user was on (and deep links as well) | M/S | ☐ |
| **AUTH-13** | Logout | Settings, then Logout | The session is cleared and going back does not open account screens | M/S | ☐ |
| **AUTH-14** | Delete account | Settings, then Delete account, then confirm | The account and its data are deleted and it can no longer be used to log in | M/I | ☐ |
| **AUTH-15** | Network error during login | Turn off the internet and try to log in | A clear error message and the ability to retry | M/S | ☐ |

## 4.3 Profile (21 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **PRF-01** | Basic info and user ID | Enter the name, photo and Bio | They are saved, and a unique user ID is generated and shown | M/I | ☐ |
| **PRF-02** | Required fields | Leave the name, Main track, Level or languages empty, then Next | Cannot continue, with a clear message (at least one language is required) | M/U | ☐ |
| **PRF-03** | Upload a profile photo | Pick a photo from the gallery or camera | The photo is uploaded and shown, and the permission is requested only when needed | M/S | ☐ |
| **PRF-04** | Write and edit the Bio | Write, then edit the Bio | The edit is saved | M/S | ☐ |
| **PRF-05** | Choose Main Track and Level | Pick a Track and a Level | The values are saved and shown in the profile | M/S | ☐ |
| **PRF-06** | Programming languages | Add a language with a level, then remove a language | Adding and removing work and the level is saved | M/S | ☐ |
| **PRF-07** | Skills with a level | Add and remove a skill with a level | Adding and removing work | M/S | ☐ |
| **PRF-08** | Soft skills and interests | Pick soft skills and interests | They are saved | M/S | ☐ |
| **PRF-09** | Other technical background | Write courses and certificates | The text is saved and shown | M/S | ☐ |
| **PRF-10** | Upload a CV | Upload a PDF under 5MB, then a non-PDF file, then a file over 5MB | The first succeeds, the second and third are rejected with a message | M/S | ☐ |
| **PRF-11** | CV permissions | Open another user's CV from a regular account, then from an Owner account after a join request | The regular account is denied, and the Owner can read applicants' CVs only (Storage rules) | M/Sec | ☐ |
| **PRF-12** | Portfolio and LinkedIn links | Add a valid link and an invalid link | The valid one is saved and the invalid one is rejected with a message | M/U | ☐ |
| **PRF-13** | Connect GitHub/Behance from Connected Accounts | Connect an account | The status shows connected only if the connection actually happened | M/S | ☐ |
| **PRF-14** | Review the Developer Snapshot | Open the Snapshot and confirm it | The summary is correct and includes the AI-Detected Skills section (if GitHub is connected) | M/S | ☐ |
| **PRF-15** | Edit the profile later | Edit data and go back to Discover | The edit is visible in the profile and the recommendations | M/I | ☐ |
| **PRF-16** | Public Profile | Open a developer's profile from another account | It is displayed to others correctly | M/S | ☐ |
| **PRF-17** | Profile completion indicator | Complete the fields one by one | The percentage updates after each field | M/S | ☐ |
| **PRF-18** | Profile Completion Banner | Use Dev-B, open Discover Home and My Profile, then dismiss the banner | The banner shows as long as the profile is incomplete (for example 60% complete) and comes back after a few days | M/S | ☐ |
| **PRF-19** | Profile Completion screen | Open the screen from the banner | A progress ring and a checklist with the missing and completed items | M/S | ☐ |
| **PRF-20** | Add button in the checklist | Tap Add on a missing item, then save | Goes to the right field, and after saving returns to the checklist with the percentage updated | M/S | ☐ |
| **PRF-21** | Setup journey (6 steps) | Move with Next and Back between the steps | Data is not lost between steps | M/S | ☐ |

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|---|---|---|---|---|---|
| **PRF-22** | Firestore users/{uid} | Inspect the document after saving | All fields (name, bio, track, level, languages, skills, ... publicId) are present and correct | M/I | ☐ |

## 4.4 GitHub Integration (8 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **GH-01** | Link the account with OAuth | Open Connect GitHub, then Connect | An explanation of the data required, OAuth with minimal scopes, and Skip works | M/I | ☐ |
| **GH-02** | Fetch repos, languages, commits and PRs | After connecting, wait for the analysis, then Re-sync again | Data comes from the Cloud Function the first time and from the cache the second time, with an animated progress | M/I | ☐ |
| **GH-03** | Skill inference | Open Review Detected Skills | Each skill comes with a confidence and evidence (languages, repos, last activity) | M/I | ☐ |
| **GH-04** | Accept and reject skills | Accept one skill and reject another | Only the accepted one is added to skills, and the status (accepted/rejected) is stored | M/I | ☐ |
| **GH-05** | GitHub Insights | Open My Profile | Top languages and activity are shown | M/S | ☐ |
| **GH-06** | Re-sync | Tap Re-sync | The data is updated and lastSync changes | M/I | ☐ |
| **GH-07** | Disconnect | Tap Disconnect, then confirm | A confirmation dialog, the analyzed data is deleted, and Insights disappear | M/I | ☐ |
| **GH-08** | Rate limit or API failure | Simulate an API failure | The user continues with manual data and a clear message (UC-01) | M/I | ☐ |
| **GH-09** | Product rules | Review the review screen and the profile | Inferred skills are in a separate section and not shown as verified facts, and there is no trusted/untrusted judgment about the person | M/S | ☐ |
| **GH-10** | Token and key security | Inspect the code and the APK | The token is in secure storage or server-side, and there are no API keys in the app (NFR-04 and NFR-07) | M/Sec | ☐ |

## 4.5 AI Match (12 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **AIM-01** | Generate recommendations | Open Discover Home with the Dev-A account | The Recommended for you section is filled using a Cloud Function + LLM | M/I | ☐ |
| **AIM-02** | List sorted by match score | Look at the order of the cards | Descending order by score and consistent with the skills | M/S | ☐ |
| **AIM-03** | The reason for each recommendation | Inspect every card | Every recommendation comes with a written reason, and no number appears alone | M/S | ☐ |
| **AIM-04** | Explain AI | Open Why this project fits you | An explanation in natural language + matched, complementary and missing skills | M/S | ☐ |
| **AIM-05** | Cross-skill match | A project needs a skill close to what the user knows (for example Kotlin while the user knows Java) | The close skill appears as a complementary skill in the explanation | M/S | ☐ |
| **AIM-06** | Skill gap | Open Skill Gap Analysis for a seed project | Missing = the project's skills minus the user's skills, with no errors | M/U | ☐ |
| **AIM-07** | Short learning tip | Look at the tip next to each missing skill | There is a short tip for each missing skill | M/S | ☐ |
| **AIM-08** | Rate the recommendation | Tap [Helpful] or [Not helpful] in the Feedback bottom sheet and write a reason for rejecting | The rating and the reason are recorded in Firestore along with promptVersion | M/I | ☐ |
| **AIM-09** | Rate explanation clarity | Rate the explanation from 1 to 5 | The rating is saved and feeds the explanation clarity metric | M/I | ☐ |
| **AIM-10** | Save to Match History | After viewing a recommendation, go back to the History | The recommendation was saved | M/I | ☐ |
| **AIM-11** | Show Match History | Open Match History | All recommendations shown before appear | M/S | ☐ |
| **AIM-12** | Fallback when the LLM is down or slow | Shut down the function or cut the internet and open the recommendations | Cached or rule-based results + a notice, and no empty screen (NFR-11) | M/I | ☐ |
| **AIM-13** | Incomplete profile | Use Dev-B and open the recommendations | Weak recommendations instead of breaking, and a "complete your profile first" message + a button that opens Profile Completion (UC-02) | M/S | ☐ |
| **AIM-14** | MatchingService | Run 8 to 10 unit tests: full match, zero match, incomplete profile, duplicate skills | Full match = highest score, zero = lowest score, and the rest without exceptions | M/U | ☐ |

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|---|---|---|---|---|---|
| **AIM-15** | Learning project for a beginner | Open the recommendations with a Beginner-level account | Learning projects are moved up in the order (project purpose is part of the matching) | M/S | ☐ |
| **AIM-16** | Guest | Tap AI Match as a Guest | The Login Required bottom sheet | M/S | ☐ |

## 4.6 Project Discovery (13 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **DSC-01** | Browse recommended projects | Open Discover Home | Recommended and Trending/Newest are shown | M/S | ☐ |
| **DSC-02** | Toggle between Projects and Developers | Tap the toggle | The list switches correctly | M/S | ☐ |
| **DSC-03** | Keyword search | Search for a word (for example Flutter) | The results contain the word, and the Search Results page appears | M/S | ☐ |
| **DSC-04** | Natural-language search | Type: I want a Flutter project for beginners | Flutter results suitable for beginners (Learning) | M/I | ☐ |
| **DSC-05** | Apply filters | Try each filter alone, then a combination: Tech stack, Difficulty, Purpose (Learning/Real), Project type, Roles needed, Team size, Track, Status (Open/Full) | The results match the filters | M/S | ☐ |
| **DSC-06** | Reset filters | Tap Reset | All filters return to their defaults | M/S | ☐ |
| **DSC-07** | Sort results | Switch between Best match and Newest | The order changes correctly | M/S | ☐ |
| **DSC-08** | Open project details | Tap a card | Project Details opens | M/S | ☐ |
| **DSC-09** | Save and Unsave | Save a project, then unsave it | The icon and the list update | M/S | ☐ |
| **DSC-10** | Express Interest | Tap Express Interest | The interest is recorded and does not turn into a join request | M/S | ☐ |
| **DSC-11** | Show Saved Projects | Open the Saved list | The saved items are shown (Saved and Favorites are one feature) | M/S | ☐ |
| **DSC-12** | Browse developers | Switch to Developers | A list of developers from the seed | M/S | ☐ |
| **DSC-13** | Developer Public Profile | Open a developer | The public profile appears | M/S | ☐ |
| **DSC-14** | Invite to project and Start chat | From the profile, tap one of them | Asks the Guest to log in, and works for a logged-in user | M/S | ☐ |
| **DSC-15** | The three actions are separate | Do Save, Express Interest and Request to Join on the same project | Each action is independent and none triggers another | M/S | ☐ |
| **DSC-16** | Empty results | Search for a word that does not exist | A clear empty state message | M/S | ☐ |
| **DSC-17** | Open and closed status | Filter Open, then Full | Full projects do not show up with Open | M/S | ☐ |
| **DSC-18** | Guest | Browse, search and filter as a Guest | Reading and search work, and actions ask for Login | M/S | ☐ |

## 4.7 Community (10 operations) (S/C)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **COM-01** | Show the Feed | Open the Community tab | Posts are shown | S/C/S | ☐ |
| **COM-02** | Create a Post | Write text + tags and publish | The post appears in the Feed | S/C/S | ☐ |
| **COM-03** | Like | Tap Like, then tap again | The counter goes up and down | S/C/S | ☐ |
| **COM-04** | Comment | Write a comment | The comment appears in Post Details | S/C/S | ☐ |

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| **COM-05** | Open a Post | Tap a post | Post details with the comments | S/C/S | ☐ |
| **COM-06** | Edit and delete your own post | Edit and delete your post, and try with someone else's post | Only your own post can be edited or deleted | S/C/Sec | ☐ |
| **COM-07** | Report | Report a piece of content | The report is recorded and a confirmation is shown | S/C/S | ☐ |
| **COM-08** | Open the author's profile | Tap the author's name | The Public Profile opens | S/C/S | ☐ |
| **COM-09** | Start Individual Chat | Tap Start Chat with the author | An individual conversation opens | S/C/S | ☐ |
| **COM-10** | Filter the Feed by tags | Pick a tag | Only the matching posts | S/C/S | ☐ |
| **COM-11** | Guest | Read as a Guest, then try Like or Comment | Reading is available and interaction asks for Login | S/C/S | ☐ |

## 4.8 Project Details (12 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **PDT-01** | Show project information | Open a project | The title, description, Tech stack, Difficulty, owner and current team are shown | M/S | ☐ |
| **PDT-02** | Purpose badge (Learning/Real) | Open a Learning project and a Real project | Learning: beginner-friendly, optional mentoring, no strict deadline. Real: commitment (hours/week), required level, deadlines, expected end date | M/S | ☐ |
| **PDT-03** | Required roles and the team | Look at the roles section | Required, open and current team are correct | M/S | ☐ |
| **PDT-04** | Match and explanation | Look at the match and Why this fits you | The match and the explanation are shown, with a link to the AI Match screen | M/S | ☐ |
| **PDT-05** | Save | Tap Save | It is saved to Saved | M/S | ☐ |
| **PDT-06** | Express Interest | Tap Express Interest | The interest is recorded | M/S | ☐ |
| **PDT-07** | Request to Join | Tap Request to Join | The request screen opens | M/S | ☐ |
| **PDT-08** | Generate a Collaboration Pitch with AI | Open the request screen | A message tailored to the project and the profile is generated | M/I | ☐ |
| **PDT-09** | Edit the message and pick the role | Edit the message and pick the required role | The edit is saved and the chosen role is sent | M/S | ☐ |
| **PDT-10** | Send the request once | Send the request, then try to send it again | The first succeeds (Request Sent), and a repeat is blocked with a message | M/I | ☐ |
| **PDT-11** | Request status | Check the request status after sending and after the owner replies | Pending, then Accepted or Rejected | M/I | ☐ |
| **PDT-12** | Share the project link | Tap Share, then open the link (Shared link) | The link opens the same project, and if not logged in it asks to Login and returns to the project | S/C/I | ☐ |
| **PDT-13** | AI failure in the pitch | Shut down the LLM and try | The user writes their own message with a clear notice | M/I | ☐ |
| **PDT-14** | Guest | Tap Save or Request as a Guest | Login, then returns to the same project | M/S | ☐ |

## 4.9 Create Project (14 operations)

Note: the AI steps (CRP-02, 03 and 08) have Should priority, and the manual path is Must.

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **CRP-01** | Choose the creation method | Open Create Entry and pick Manual or AI-assisted | Both lead to the right creation screens | M/S | ☐ |
| **CRP-02** | Describe the idea | Write the idea in two sentences (Describe Your Idea) | The description is accepted and sent to the AI | S/C/S | ☐ |
| **CRP-03** | Generate the setup with AI | Tap Generate in AI Project Setup | Description, Tech stack, roles, difficulty and suggested tasks, all editable | S/C/I | ☐ |
| **CRP-04** | Edit the title and description | Edit in the Project Details Form | The edit is saved | M/S | ☐ |
| **CRP-05** | Choose the project type | Pick Learning, then Real | The related fields appear: commitment hours, minimum level, Mentoring, expected end date | M/S | ☐ |
| **CRP-06** | Choose and edit the Tech stack | Add and remove technologies | The list updates | M/S | ☐ |

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|---|---|---|---|---|---|
| **CRP-07** | Confirm the required skills | Review the Required Skills and confirm them | They are saved with the project | M/S | ☐ |
| **CRP-08** | Generate roles with AI | Tap Generate in Role Generator | Suitable roles (Frontend, Backend, QA...) | S/C/I | ☐ |
| **CRP-09** | Edit, add and delete roles | Edit, add and delete | Saved correctly, and a project cannot have no role at all | M/S | ☐ |
| **CRP-10** | Team size, Difficulty and Project type | Set the values | They are shown in Review and Publish | M/S | ☐ |
| **CRP-11** | Save a Draft | Tap Save Draft, close the app and open it again | The Draft exists for the owner only and is not shown in Discover | M/S | ☐ |
| **CRP-12** | Publish with owner approval | Tap Publish | The project appears in Discover, and nothing is published automatically without the owner's tap | M/I | ☐ |
| **CRP-13** | Review and Publish | Open the review screen | A clear summary of the type (Learning/Real) before publishing | M/S | ☐ |
| **CRP-14** | Publish Success and Missing Roles | After publishing, tap Find Missing Roles | Missing Role Alert and Find Suitable Developers work | M/S | ☐ |
| **CRP-15** | AI failure | Shut down the LLM and try the AI path | Manual input with a clear message | M/I | ☐ |
| **CRP-16** | Validation | Leave the title or description empty and publish | Error messages and no publishing | M/U | ☐ |
| **CRP-17** | Guest | Tap Create Project as a Guest | Login Required, then back to the same screen | M/S | ☐ |

## 4.10 My Teams (12 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **TEM-01** | Show the teams list | Open My Teams with a Member account | Only the teams they are a member of | M/S | ☐ |
| **TEM-02** | Open the Workspace | Pick a team | Project Info, Members, Team Chat and Tasks are shown | M/S | ☐ |
| **TEM-03** | Members and their roles | Open Team Members | The members, their roles and the missing roles are correct | M/S | ☐ |
| **TEM-04** | Real-time Team Chat | Two accounts on two devices, send a message | The message arrives instantly on the other device | S/C/I | ☐ |
| **TEM-05** | Create a Task | Write a description, pick an assignee, deadline and priority, and Save | The Task appears in the Task Board | M/S | ☐ |
| **TEM-06** | Assign a Task to a member | Change the assignee | It appears for the new member | M/S | ☐ |
| **TEM-07** | Update a Task's status | Move it between To Do, In Progress, In Review and Done | The status updates for the whole team | M/I | ☐ |
| **TEM-08** | Edit and delete a Task | Edit and delete | The change is applied after confirming the delete | M/S | ☐ |
| **TEM-09** | Filter Tasks | Filter by assignee, then by status | The results match | M/S | ☐ |
| **TEM-10** | Missing Role Alert inside the team | Open a team that is missing a role | An alert for the missing role and a button to recommend people | M/S | ☐ |
| **TEM-11** | Leave the team | Tap Leave Team and confirm | The member leaves and the team disappears from their list | M/I | ☐ |
| **TEM-12** | FCM notification for the Task assignee | Assign or update a Task and watch the other device (background and terminated) | The notification arrives in 10 seconds or less (NFR-02) | M/I | ☐ |
| **TEM-13** | Task Details | Open a Task | The description, assignee, status, deadline and priority are shown | M/S | ☐ |
| **TEM-14** | Offline | Turn off the internet and create a Task | The operation is queued or the user is asked to connect (UC-05), and it is not lost | M/I | ☐ |
| **TEM-15** | Team permissions | A user who is not a member tries to open a team Workspace | Denied (Firestore rules) | M/Sec | ☐ |
| **TEM-16** | Loading, empty and error states | Open the Workspace on a slow connection and with a team that has no Tasks | Clear states and no empty or blank screen | M/S | ☐ |

## 4.11 My Projects (13 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **PRJ-01** | Navigate between Ongoing, Invited and Previous | Switch between the tabs | Each project is in the right tab according to its status and role | M/S | ☐ |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **PRJ-02** | Open Project Overview (Owner) | Open a project you own | The status, requests, invitations and editing are shown | M/S | ☐ |
| **PRJ-03** | Edit the project | Edit the data | The edit is saved and shown in Discover | M/S | ☐ |
| **PRJ-04** | Close and archive the project | Tap Close, then Archive | The project goes to Previous and does not show in Discover | M/S | ☐ |
| **PRJ-05** | Show incoming requests | Open Incoming Requests | The pending requests are shown | M/S | ☐ |
| **PRJ-06** | Accept a request | Accept a request from an Applicant | The applicant is added to the team and gets a notification, and the owner is confirmed too | M/I | ☐ |
| **PRJ-07** | Reject a request | Reject a request | A notification to the applicant and the request is closed | M/I | ☐ |
| **PRJ-08** | Applicant details | Open Applicant Details | The profile, matched skills, the applicant's message and the CV | M/S | ☐ |
| **PRJ-09** | Invite a developer | Tap Invite Developer and pick a developer | The invitation is sent and a notification reaches them | M/I | ☐ |
| **PRJ-10** | Accept and decline an invitation | From the invitee's account, accept once and decline once | Accepting adds them to the team, and declining closes the invitation | M/I | ☐ |
| **PRJ-11** | Cancel an invitation | From the owner's account, cancel a pending invitation | The invitation is cancelled and the invitee sees that it was cancelled | M/I | ☐ |
| **PRJ-12** | Change the project status | Switch between Open, Full and Completed | The status updates and affects the filters and applying | M/S | ☐ |
| **PRJ-13** | FCM notifications | Perform these events one by one (new request, acceptance, rejection, invitation) | Each notification arrives in 10 seconds or less | M/I | ☐ |
| **PRJ-14** | Owner and Member role at the project level | The same account is an owner of one project and a member of another | Permissions differ per project | M/S | ☐ |
| **PRJ-15** | Owner permissions | A regular member tries to view Incoming Requests or edit the project | Denied | M/Sec | ☐ |
| **PRJ-16** | Invite a user who is not registered (S/C) | Open the invitation link with a new account | Logs in or creates an account and returns to the same project (US-17) | S/C/I | ☐ |

## 4.12 Prompt Evaluation (7 operations)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **PEV-01** | Rate an AI output | After a recommendation, a project setup or a pitch, open Evaluate AI Suggestion | The rating screen appears: [Helpful]/[Not helpful], clarity from 1 to 5 and an optional reason | M/S | ☐ |
| **PEV-02** | Store the rating with the version number | Rate an output from version v1, then v2 | The rating is stored in Firestore with promptVersion | M/I | ☐ |
| **PEV-03** | Compare prompt versions | Open the Prompt Evaluation Report | A table with the number of ratings, the helpful rate and the average explanation clarity for each version | M/S | ☐ |
| **PEV-04** | Compute the metrics | Compare the table's numbers with the raw data | Relevance and explanation clarity match the manual calculation | M/U | ☐ |
| **PEV-05** | Run the versions on a fixed test set | Run each version on a fixed set (for example 5 profiles × 10 projects) | The results are recorded in a table inside the documentation (this is the evidence required for submission) | M/S | ☐ |
| **PEV-06** | Activate the best version | Change activeVersion (Remote Config or the Firestore config) | The app uses the new version without an app update | M/I | ☐ |
| **PEV-07** | Ratings permissions | A user tries to write or read another user's rating | Denied | M/Sec | ☐ |
| **PEV-08** | The report screen is internal | Open the report with a regular account, then with the demo account | The report is for developers and the demo only | M/S | ☐ |

## 4.13 Chatbot (9 operations) (Could, locked in Free as a Pro feature)

| ID | Operation | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **BOT-01** | Open the chat | From the floating icon or from Discover or Profile | The Chatbot screen opens | S/C/S | ☐ |
| **BOT-02** | Send a question and receive a reply | Type a question and send | An understandable and appropriate reply | S/C/I | ☐ |
| **BOT-03** | Suggested prompts | Tap a suggestion (What skills am I missing? / Recommend a project for me to learn) | The suggestion is sent and the reply is shown | S/C/S | ☐ |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **BOT-04** | Context-based replies | Ask about your skills | The reply is based on the profile and skills | S/C/I | ☐ |
| **BOT-05** | Deep link from the reply | Tap a link to a project or a screen | Opens the right destination | S/C/S | ☐ |
| **BOT-06** | Clear the conversation history | Tap Clear | The conversation is cleared | S/C/S | ☐ |
| **BOT-07** | Request cap | Send more requests than the limit | A clear message about the maximum | S/C/I | ☐ |
| **BOT-08** | Fallback when the LLM is down | Shut down the LLM and send a question | A clear message and no empty screen | S/C/I | ☐ |
| **BOT-09** | Rate the reply | Tap [Helpful] or [Not helpful] on a reply | The rating is recorded | S/C/I | ☐ |
| **BOT-10** | Access by plan | Open it with a Free account, then Pro, then Guest | Free: the upgrade information screen. Pro: works. Guest: Login Required | S/C/S | ☐ |

## 5. Testing the Free and Pro Flavors

| ID | Case | Steps | Expected | Priority/Type | Result |
|---|---|---|---|---|---|
| **FLV-01** | Build both flavors | Build free and pro from the same code | Both flavors build, install and run | M/S | ☐ |
| **FLV-02** | Free features | Run Free | Basic discovery, matching, project profile and Community are available | M/S | ☐ |
| **FLV-03** | Pro features | Run Pro | The features locked in Free (such as Chatbot and AI insights) work | M/S | ☐ |
| **FLV-04** | Smart-feature request flow | In Free, request a locked feature | The system checks the plan: if available it works, if not the upgrade information screen appears | M/S | ☐ |
| **FLV-05** | Upgrade decision | From the upgrade screen, tap continue, then go back | Either continues to the upgrade confirmation or returns to the available features | M/S | ☐ |
| **FLV-06** | No invented prices or limits | Review the plan screens (Plan comparison and Upgrade confirmation) | No prices or usage limits are written, "Plan benefits to be confirmed" is present, and there is no payment processing | M/S | ☐ |
| **FLV-07** | Google and FCM in release | Try Google Sign-in and notifications on the release build for each flavor | Both work (the two things most likely to break in release) | M/I | ☐ |
| **FLV-08** | Additional screens | Open Match History, Plan comparison and Upgrade confirmation | Navigation works correctly | M/S | ☐ |

## 6. Non-Functional Requirements (NFR) Testing

| ID | Requirement | Measurement method | Acceptance criterion | Result |
|---|---|---|---|---|
| **NFR-01** | Recommendation loading speed | Timing measurement, 10 runs on a real device | 5 seconds or less | ☐ |
| **NFR-02** | Notification delivery speed | FCM in foreground, background and terminated on two real devices | 10 seconds or less | ☐ |
| **NFR-03** | Firestore security rules | Test on the Emulator for every collection: read and write with the data owner's account, another account and a Guest | 100% of the checklist passes, and a user reaches only their own data | ☐ |
| **NFR-04** | No keys in the code | Search the repo and the APK after decompiling the build | Zero secrets, and the GitHub and LLM keys are in Cloud Functions only | ☐ |
| **NFR-05** | Runtime permissions | Review all permission requests (photos, camera, notifications) | They are requested only when needed | ☐ |
| **NFR-06** | Code obfuscation | Build an AAB with --obfuscate and --split-debug-info and inspect it | Names are unreadable and the app works | ☐ |
| **NFR-07** | GitHub privacy | Review the scopes and the stored data and test Disconnect | Only data needed for matching, and the disconnect option works | ☐ |
| **NFR-08** | Usability | 10 testers perform: login, finding a project, sending a request, without help | The core tasks are completed without help | ☐ |
| **NFR-09** | Consistent UI and animation | Review the screens in Light and Dark | A shared theme and widget kit and smooth animation | ☐ |
| **NFR-10** | Stability | Crashlytics on the testers' sessions + cutting the internet | 95% of sessions without a crash, and clear network error messages | ☐ |
| **NFR-11** | Graceful AI degradation | Shut down the LLM in all AI features (AIM-12, PDT-13, CRP-15 and BOT-08) | Fallback results or a message, and no empty screen | ☐ |
| **NFR-12** | Compatibility | Run on two real devices with different versions and sizes | No overflow or crash | ☐ |

| | | | | |
|---|---|---|---|---|
| **NFR-13** | Git workflow | Review the PRs and branches on GitHub | 100% of merges were reviewed through a PR, and use feature/module-name | ☐ |
| **NFR-14** | Documentation | Review the README and the technical documentation at the last milestone | Complete and following best practices | ☐ |

# 7. End-to-End (E2E) Testing and KPIs

## 7.1 E2E Scenarios

We test them on two real devices with the release build: Device A for the developer and Device B for the project owner.

| ID | Scenario | Steps | Expected | Priority | Result |
|---|---|---|---|---|---|
| **E2E-01** | The core MVP flow | Splash, then Login, then Profile, then GitHub import, then Discover, then Project Details (Learning/Real), then Skill Gap, then Request to Join, then an FCM notification to the owner, then the owner accepts, then Team Workspace and Tasks, then Prompt Evaluation | The entire flow works end to end without a crash (must be finished before 30 November) | M | ☐ |
| **E2E-02** | The owner flow | Create Project (manual, then AI), then Publish, then Missing Role Alert, then Find Suitable Developers, then Invite, then the invitation is accepted | The team is formed and the roles are updated | M | ☐ |
| **E2E-03** | The Guest flow | Browse, then an action that needs an account, then Login, then return to the same screen | The flow is correct | M | ☐ |
| **E2E-04** | Notifications and Deep Links | Open a notification or a link (project, invitation, request, message, Task update) with a logged-in account, a logged-out one, and a cancelled invitation | The right destination; if logged out, Login and then restoring the destination; and a cancelled invitation shows an appropriate message | M | ☐ |
| **E2E-05** | Feedback flow | Open a recommendation and rate it [Helpful] and then [Not helpful] | The rating is recorded and available for quality assessment | M | ☐ |
| **E2E-06** | AI access by plan | Request a smart feature in Free and in Pro | According to FLV-04 | M | ☐ |

## 7.2 KPI Measurement (10 testers and a feedback form)

| KPI | Target | Measurement method |
|---|---|---|
| Recommendation response time | 5 seconds or less | Timing measurement on a real device |
| FCM notification delivery | 10 seconds or less | Measurement on two devices |
| Crash-free sessions | 95% or more | Crashlytics |
| Share of relevant recommendations | 60% or more | Testers' rating (relevant or not) |
| Average explanation clarity | 4 out of 5 or more | Testers' form |
| Profile completion | 80% of testers complete it | Observation during testing |

# 8. Coverage Summary

| Module | Operations in Core Features | Number of Test Cases | Priority |
|---|---|---|---|
| 1. Splash and Onboarding | 6 | 7 | Must |
| 2. Authentication (with Guest) | 10 | 15 | Must |
| 3. Profile | 21 | 22 | Must |

| | | | |
|---|---|---|---|
| 4. GitHub Integration | 8 | 10 | Must |
| 5. AI Match | 12 | 16 | Must |
| 6. Project Discovery | 13 | 18 | Must |
| 7. Community | 10 | 11 | Should/Could |
| 8. Project Details | 12 | 14 | Must |
| 9. Create Project | 14 | 17 | Must (AI assist: Should) |
| 10. My Teams | 12 | 16 | Tasks: Must, Chat: Should |
| 11. My Projects | 13 | 16 | Must |
| 12. Prompt Evaluation | 7 | 8 | Must |
| 13. Chatbot | 9 | 10 | Could |
| Free/Pro | — | 8 | Must |
| NFR-01 to NFR-14 | — | 14 | Must |
| E2E | — | 6 | Must |
| Total | 147 | 208 | |

The number of Test Cases is larger than the number of operations because some operations have more than one case (a success case, an error case and permissions).

## 9. Execution Log and Sign-off

### 9.1 Execution Rounds Log

| Round | Date | Build | Devices | Passed | Failed | Blocked | Tester |
|---|---|---|---|---|---|---|---|
| QA Round 1 | 19 to 20 November | | | | | | |
| KPIs with testers | 20 to 28 November | | | | | | |
| QA Round 2 (regression) | 24 to 28 November | | | | | | |

### 9.2 Sign-off

| Role | Name | Date | Signature |
|---|---|---|---|
| QA | Hany Rana | | |
| Project Manager | Mohraeel | | |

## 10. Open Points Affecting Testing

- Chatbot, Community and Chat: if they are postponed to future work, their Test Cases (COM, BOT and TEM-04) are removed from the scope of the round.
- My Teams and My Projects: if they are merged into one tab, the Test Cases stay as they are and only the navigation changes.
- The CV: testing currently covers upload, display and permissions only (AI analysis is not in the plan).
- Behance: testing currently covers linking only (PRF-13), without data analysis.
- Project Purpose (Learning/Real): it is not yet an official FR in the requirements, so we relied on the Core Features document, and the tests (PDT-02, CRP-05 and AIM-15) must be reconfirmed once it is approved.
- Cloud Functions need Firebase Blaze, which must be confirmed before the first QA round.

- The match score weights are just an example in the ideas list, so the AIM-02 test checks the order and the logic rather than fixed numbers until the team approves the weights.